Amplitude Modulation and Demodulation

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1 Introduction

When data is transmitted over long distances, a higher frequency is often preferred in many cases. A higher frequency allows smaller antennas and transmission over further distances. However, most signals that would be transmitted typically do not have these high frequencies. One method to solve this is with amplitude modulation (AM). Amplitude modulation is a method in which a high frequency wave, often called the carrier, is multiplied by the signal containing the information, the modulation signal, you wish to transmit. This will result in a signal with the amplitude of the modulation signal and the frequency of the carrier. To represent this modulation technique mathematically, the following equation can be used.

$$A[1 + cos(\omega_m t)]cos(\omega_c t) \tag{1}$$

When expanded, the resulting equation can be obtained.

$$Acos(\omega_c t) + \frac{A}{2}cos((\omega_c + \omega_m)t) + \frac{A}{2}cos((\omega_c - \omega_m)t) \tag{2}$$

Through this expansion, it can be seen that an AM signal will have one primary center frequency and two sideband frequencies. Recalling that all information is stored in the modulation signal, the information being transmitted is purely stored in the two sideband frequencies. It should also be noted that AM spreads its power between 3 frequencies, making it more prone to noise and less bandwidth efficient than other modulation techniques. These are some of the trade-offs for the simplicity and wide range AM offers.

The purpose of increasing the frequency of our modulating signal is to allow it to travel further distances and for other signals to be transmitted with little to no interference from one another by changing the carrier frequency. However, when the signal reaches the receiver, it will need to be demodulated to reconstruct the original information. To reconstruct our signal, it will be sent through a rectifier circuit and a low pass filter.

2 Data and Analysis

To simulate amplitude modulation using the TIMS machine, the schematic shown in Figure 1 can be used.

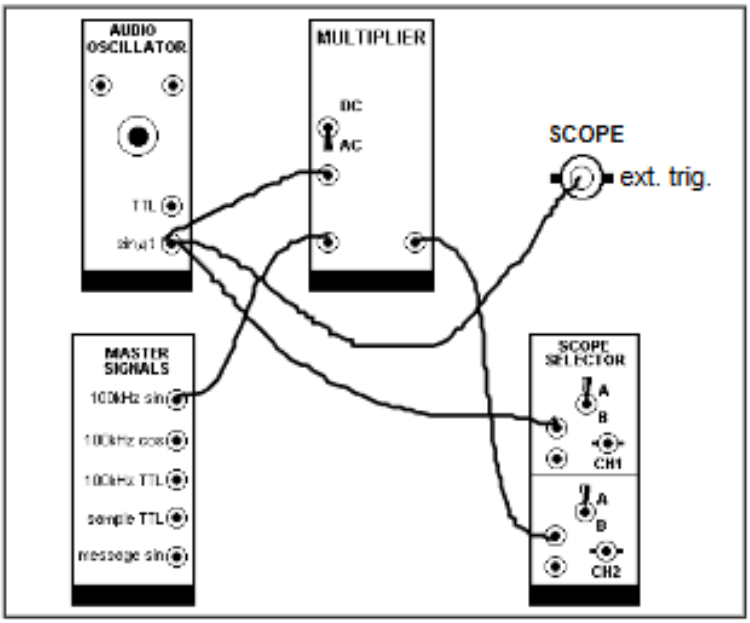


Figure 1: Connection schematic for amplitude modulation using the TIMS device

For this experiment, the sine wave will be set to 2 [kHz] and 2 [V_{pp}] with a DC offset of 1 [V]. The carrier frequency, generated by the Master Signal module, will be set to 100 [kHz]. The resulting wave will look much like Figure 2 below.

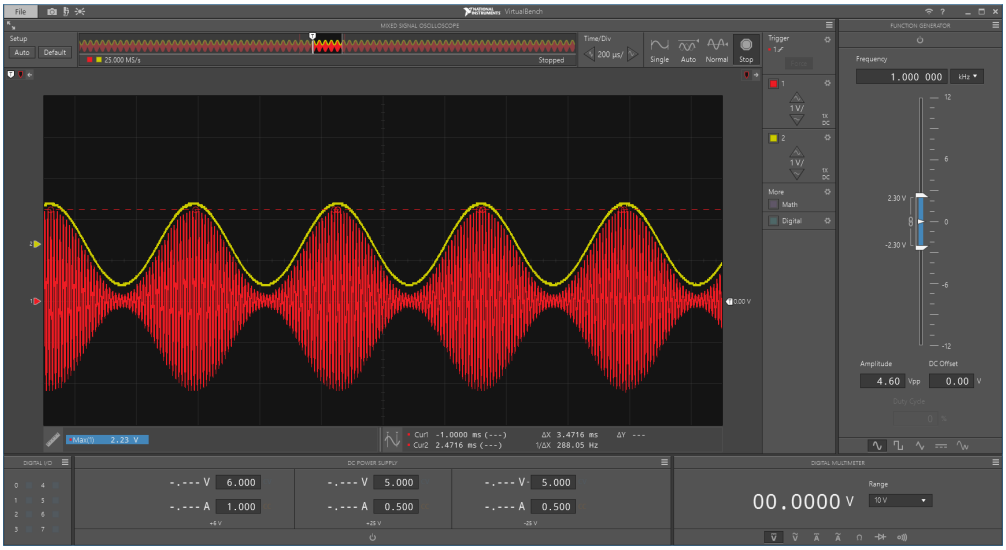


Figure 2: Original signal in yellow and the amplitude modulated signal in red

Next, the carrier will be generated with a voltage controlled oscillator at a much lower frequency of 17 [kHz]. The wave will look much like that in Figure 2 with a lower frequency. This can be seen in Figure 3.

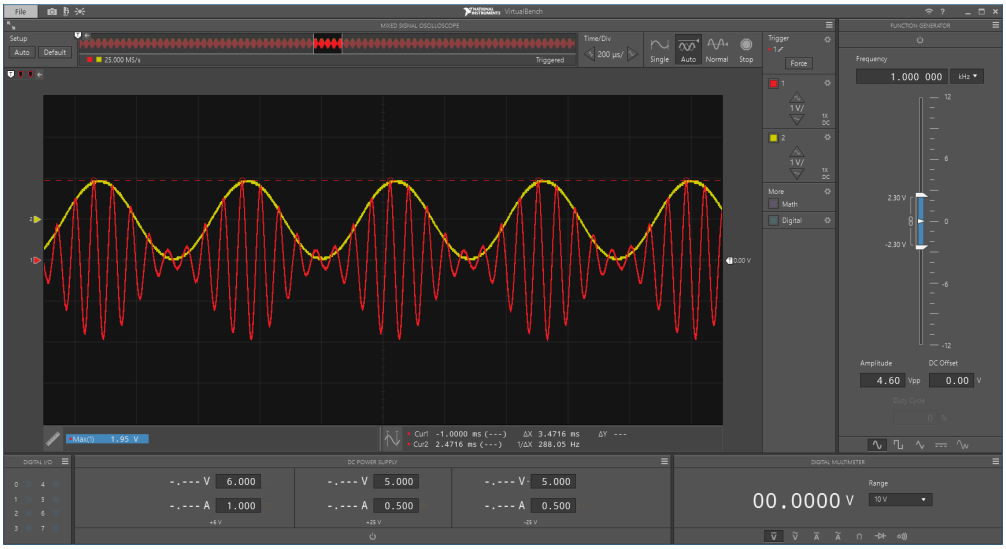


Figure 3: Original signal in yellow and the amplitude modulated signal in red

The lower frequency of the carrier wave allows us to clearly see how the frequency of the carrier wave being mapped to the amplitude of the original signal. Figure 4 shows the frequency spectrum of the modulated signal.

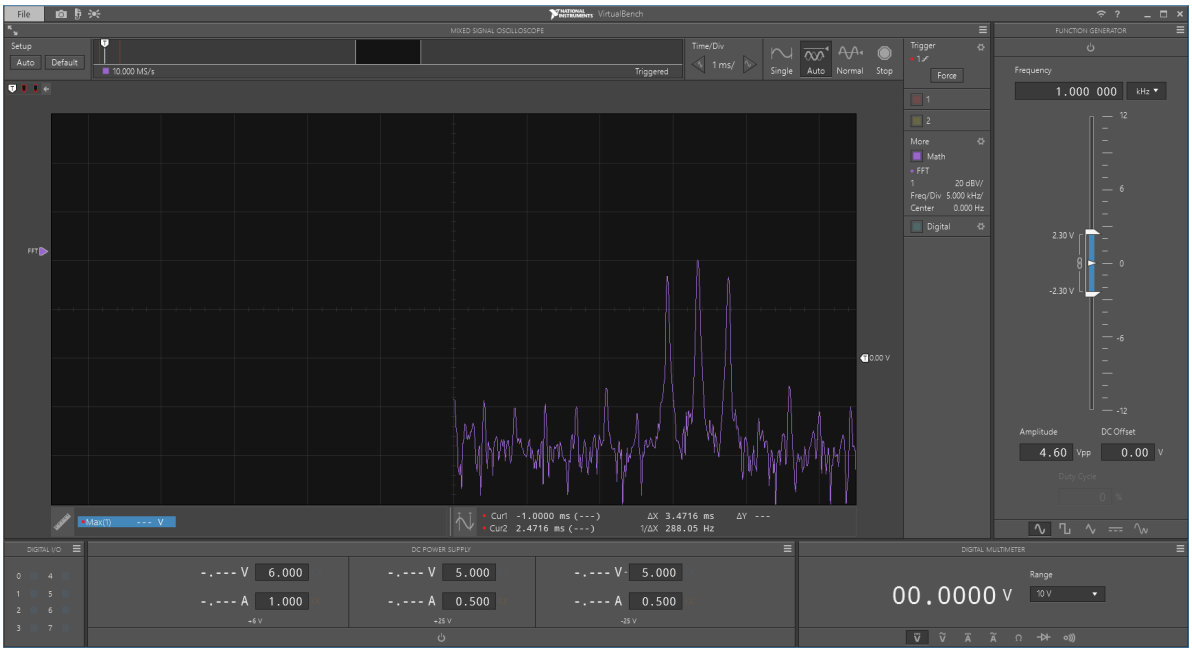


Figure 4: Frequency spectrum of modulated signal

A central spike can be seen at 17 [kHz] with two sidebands at 15 [kHz] and 19 [kHz], which is the expected shift. In order to demodulate the signal, it will need to pass through a rectifier and a low pass filter. The resulting wave is seen in Figure 5.

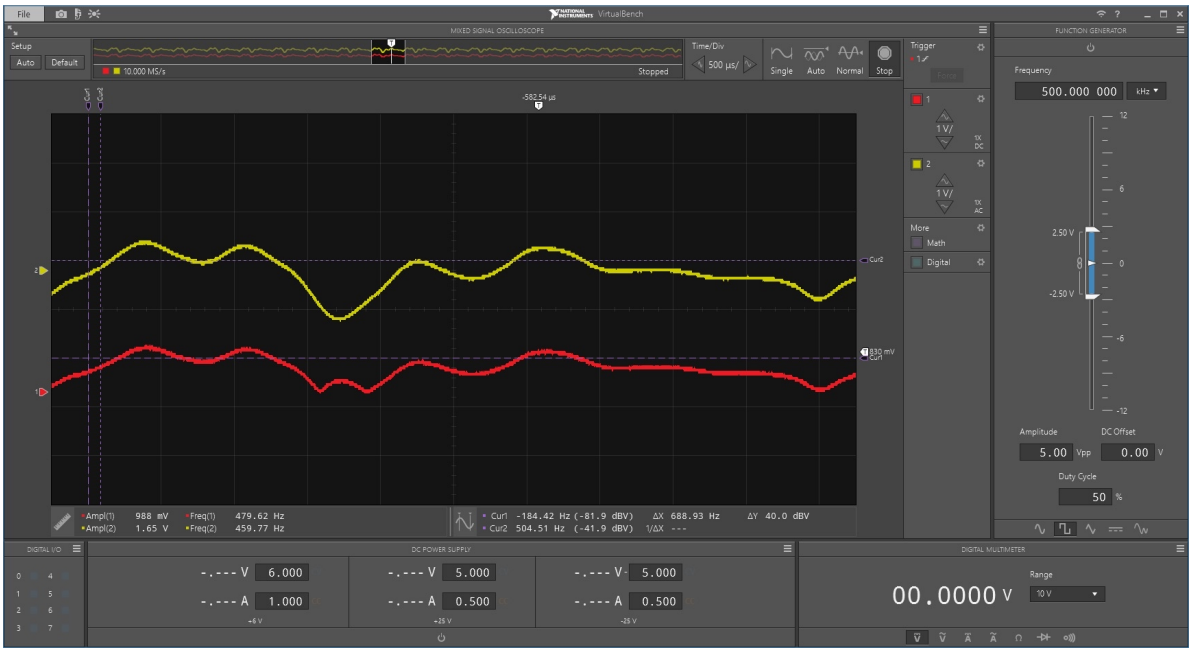


Figure 5: Original signal in yellow and the demodulated signal in red

The image shows the combination of the rectifier and low pass filter successfully reconstructed the original signal. It is important to consider the limitations of amplitude modulation. When the envelope becomes bigger than the DC offset, overmodulation will occur. This will cause more sidebands to form as they will no longer hold a faithful copy of the original signal. Figure 6 shows what happens when overmodulation occurs in the frequency spectrum.

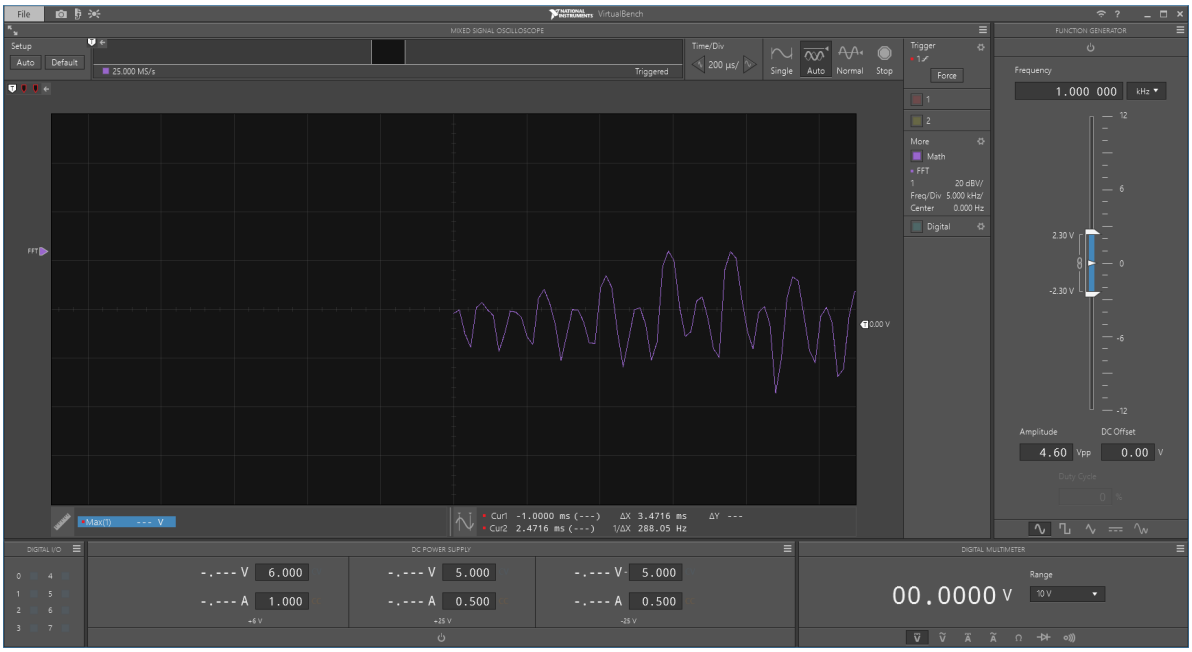


Figure 6: Overmodulated signals frequency spectrum

3 Tutorial Questions

1. There is no difficulty in relating the formula of eqn. (5) to the waveforms of Figure 7 for values of 'm' less than unity. But the formula is also valid for m $>$ 1, provided the magnitudes P and Q are interpreted correctly. By varying 'm', and watching the waveform, can you see how P and Q are defined for m $>$ 1?

   Answer: For m>1, the classical formula for AM is still used, but Q becomes negative at points, causing phase reversals and envelope distortion. P and Q are still mathematically determined as $A_c(1+m)$ and $A_c(1-m)$, but for m>1, Q is interpreted as a negative amplitude meaning distortion is present and the waveform will show broken, reversed segments.

2. Derive eqn.(5), which relates the magnitude of the parameter 'm' to the peak-to-peak and trough-to-trough amplitudes of the AM signal.

   Answer: For some AM signal $s(t) = A_c[1 + mcos(\omega_m t)]cos(\omega_c t)$ the envelope $E(t)$ is $E(t) = A_c|1 + mcos(\omega_m t)|$. We can then define the peak amplitude $P = A_c(1+m)$ and the trough amplitude $Q = A_c(1-m)$. If we then add and subtract P and Q, we find $P + Q = A_c(1+m) + A_c(1-m) = 2A_c$ and $P - Q = A_c(1+m) - A_c(1-m) = 2A_c m$ Then we simply divide the two equations to derive our final solution

$$\frac{P-Q}{P+Q} = \frac{2A_c m}{2A_c} = m$$

3. If the AC/DC switch on the MULTIPLIER front panel is switched to AC what will the output of the model of Figure 6 become?

   Answer: If there is no longer a DC component to our signal, the waveform will be centered at 0 and likely overmodulated unless the carrier amplitude is equal to or greater then 1 [V].

4. An AM signal, depth of modulation 100% from a single tone message, has a peak-to-peak amplitude of 4 volts. What would an RMS voltmeter read if connected to this signal? You can check your answer if you have a WIDEBAND TRUE RMS METER module.

   Answer: For a fully modulated AM wave, the RMS voltage is $V_{rms} = V_c\sqrt{1 + \frac{m^2}{2}}$. For a wave with $V_{pp} = 4[V]$, $V_c = \frac{V_{pp}}{2} = 2[V]$ giving us a final value of $V_{rms} = 1.732[V]$. This means an RMS voltmeter will read 1.73 [V] for a fully modulated AM signal with $4[V_{pp}]$ amplitude.

5. In Task T6, when modelling AM, what difference would there have been to the AM from the MULTIPLIER if the opposite polarity ($+V_e$) had been taken from the VARIABLE DC module?

Answer: If we used the opposite polarity, $+V_e$, the new DC offset would keep the AM envelope from tipping over downward and help prevent overmodulation on the negative half which would result in a more positive, upright signal envelope.

Certification: **I hereby certify that performance of the experiment above and the writing of this report was entirely by me (or by me and my lab partner). I understand that if this certi fication is false, I am in violation of Academic Honesty rules and may be subject to serious penalty in accordance with those rules.**

Signature Josiah Germain **Date** 11/20/25